ARTIFICIAL INTELLIGENCE

Study Notes — Chapters 1 & 2

Based on: Artificial Intelligence: A Modern Approach, 4th Edition

Russell & Norvig

Program: ADP (Associate Degree Program)	**Course:** Introduction to AI
Topics: Introduction, History, Components, Agents	**Coverage:** AIMA 4e Ch.1 & Ch.2

CHAPTER 1 — INTRODUCTION TO ARTIFICIAL INTELLIGENCE

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is the branch of computer science concerned with building machines that **think and act intelligently**. It covers everything from learning and reasoning to language, robotics, games, and medicine.

> **Simple Definition: AI = making computers do things that, when done by humans, would require intelligence.**

1.1 Four Approaches to AI

AI researchers have followed two dimensions: Human vs. Rational, and Thought vs. Behavior. These give four approaches:

Approach	Goal	Key Idea	Example
Acting Humanly	Mimic human behaviour	Turing Test — fool a human judge	Chatbots, ChatGPT
Thinking Humanly	Mimic human thought	Cognitive modelling — match reasoning steps	GPS program by Newell & Simon
Thinking Rationally	Reason correctly	Laws of thought / formal logic	Prolog, theorem provers
Acting Rationally	Act to achieve the best outcome	Rational agent — maximise performance	Self-driving cars, chess engines

1.2 The Turing Test (Acting Humanly)

Proposed by **Alan Turing (1950)**. A computer **passes** if a human interrogator cannot tell whether responses come from a person or a machine.

Turing Test — Required Capabilities

Standard Turing Test:

- Natural Language Processing (NLP) — understand and generate language
- Knowledge Representation — store facts about the world
- Automated Reasoning — draw conclusions from stored knowledge
- Machine Learning — adapt to new situations

Total Turing Test (also needs):

- Computer Vision — perceive objects
- Speech Recognition — hear and understand speech
- Robotics — physically interact with the world

1.3 The Rational Agent Approach (Standard Model)

Rational Agent: An entity that acts to achieve the **best expected outcome** given what it knows. This is the **standard model** of modern AI — more general and more scientific than simply trying to copy human behaviour.

> **Rationality ≠ Perfection. A rational agent does the best it can with available information — it does not need to be omniscient.**

1.4 The Value Alignment Problem

If we cannot perfectly specify our objectives to a machine, the machine may achieve exactly what we asked for but NOT what we actually wanted — this is the King Midas Problem.

Problem	Description	Solution Direction
Value Alignment	Machine objectives do not match human values	Let machines learn preferences
King Midas Problem	Machine does exactly what asked, not what intended	Give machines uncertainty about goals
Control Problem	Controlling machines smarter than us	Allow humans to switch off AI
Gorilla Problem	Humans may lose control to super-AI	Design provably beneficial AI

Provably Beneficial AI — New Standard Model

Instead of fixed objectives, AI machines should:

- Be uncertain about human objectives
- Learn human preferences from observation
- Defer to human control
- Allow themselves to be switched off

This leads to: **Assistance Games**, **Inverse Reinforcement Learning**, and **Provably Beneficial AI**

CHAPTER 1 — HISTORY & FOUNDATIONS OF AI

2. Foundations of AI

AI draws from many disciplines. Each field contributes essential ideas:

Field	Key Contribution to AI
Philosophy	Can machines reason? How does mind arise from brain?
Mathematics	Formal logic, probability theory, complexity (NP-completeness)
Economics	Decision theory, utility functions, game theory, MDPs
Neuroscience	How brains process information; neural circuits; fMRI
Psychology	Behaviorism and cognitive psychology; how humans think
Computer Engineering	Fast hardware; Moore's law; GPUs and TPUs for deep learning
Control Theory	Self-regulating machines; feedback loops; stochastic control
Linguistics	Language and thought; NLP; computational linguistics

3. History of AI — Timeline

Period	Name	Key Events
1943–1956	Inception	McCulloch & Pitts neurons (1943); Turing's paper (1950); SNARC neural computer; Dartmouth Workshop (1956) — AI named
1952–1969	Early Enthusiasm	GPS by Newell & Simon; Lisp; checkers-playing program by Samuel; perceptrons
1966–1973	First AI Winter	Overconfidence exposed; Lighthill report; Minsky & Papert's Perceptrons; funding cut
1969–1986	Expert Systems Era	DENDRAL, MYCIN, XCON; Prolog; Fifth Generation project; $1B+ industry
1986–Present	Neural Networks Return	Backpropagation reinvented; PDP/connectionism books
1987–Present	Probabilistic Reasoning	Bayesian networks; HMMs; speech recognition; TD-Gammon
2001–Present	Big Data	Web-scale data; ImageNet challenge; Watson on Jeopardy!
2011–Present	Deep Learning Era	AlexNet (2012); AlphaGo; AlphaZero; GPT; DALL-E; ChatGPT

Key Turing Award Winners in AI: Minsky (1969), McCarthy (1971), Feigenbaum & Reddy (1994), Judea Pearl (2011), Hinton, LeCun, Bengio (2019)

CHAPTER 2 — INTELLIGENT AGENTS

4. What is an Agent?

An **agent** is anything that perceives its **environment** through **sensors** and acts on it through **actuators**. A human, robot, thermostat, and software bot are all agents.

4.1 Agent Anatomy Diagram

ENVIRONMENT		
Percepts →	**AGENT** *Architecture + Program* Agent Function: percept sequence → action	→ Actions

4.2 Key Terms

Percept	The input received by the agent at any given moment (e.g., camera image, temperature reading)
Percept Sequence	The complete history of all inputs the agent has received
Agent Function	The mapping from percept sequences to actions — defines ideal behaviour
Agent Program	The concrete code or algorithm that implements the agent function
Agent Architecture	The hardware platform: computing device + sensors + actuators
Actuators	Components through which the agent acts (e.g., wheels, speakers, API calls)

5. Rational Agents

A **rational agent** selects actions expected to **maximise its performance measure** given: (1) the performance measure, (2) prior knowledge, (3) available actions, and (4) percept sequence to date.

Properties of a Rational Agent
• Gathers information actively (does not wait passively) • Learns from experience to improve future performance • Is autonomous — does not rely solely on built-in knowledge • Maximises expected performance, not guaranteed perfect performance

6. PEAS Framework — Designing an Agent

Before building an agent we must specify its task environment using **PEAS**:

Letter	Stands For	Question to Ask
P	Performance Measure	How do we judge success? (e.g., safety, time, cost)
E	Environment	Where does the agent operate? (road, internet, office)
A	Actuators	How does the agent act? (steering, keyboard, robot arm)
S	Sensors	How does the agent perceive? (camera, GPS, keyboard input)

6.1 PEAS Example — Automated Taxi

PEAS Component	Details for Automated Taxi
Performance	Safe trip, fast delivery, legal driving, passenger comfort, profit
Environment	Roads, traffic, pedestrians, weather, other vehicles, traffic signs
Actuators	Steering wheel, accelerator, brake, horn, lights, indicators
Sensors	Cameras, GPS, speedometer, lidar, radar, sonar, microphone

7. Types of Task Environments

The nature of the environment determines the right type of agent. Every environment can be described on seven dimensions:

Property	Options	What it Means	Example
Observability	Fully / Partially	Can sensors give the agent the complete state?	Chess (full) vs Poker (partial)

Property	Options	What it Means	Example
Agents	Single / Multi	Is there one agent or several interacting?	Crossword (single) vs Chess (multi)
Determinism	Deterministic / Stochastic	Is the next state fully determined by current?	Chess (det.) vs Taxi (stoch.)
Episodicity	Episodic / Sequential	Does current decision affect future decisions?	Part-picking (ep.) vs Chess (seq.)
Dynamics	Static / Dynamic	Does environment change while agent deliberates?	Crossword (static) vs Taxi (dynamic)
Time	Discrete / Continuous	Are time, state, and actions discrete or continuous?	Chess (disc.) vs Driving (cont.)
Knowledge	Known / Unknown	Are the rules of the environment known upfront?	Chess (known) vs new video game (unknown)

Hardest Environment: Partially observable + Multiagent + Stochastic + Sequential + Dynamic + Continuous + Unknown.

CHAPTER 2 — AGENT STRUCTURES & ARCHITECTURES

8. Five Types of Agent Programs

Simple Reflex Agent

Acts only on the current percept. Uses condition-action rules (IF state THEN action). Fast but cannot handle partial observability.
Example: Thermostat (IF temp < 18°C THEN turn on heater)

Model-Based Reflex Agent

Maintains an internal state model of the world. Uses a transition model (how the world evolves) and a sensor model (what percepts mean).
Example: Robot vacuum that remembers which areas it has cleaned

Goal-Based Agent

Has explicit goals. Considers the future. Chooses actions that bring it closer to achieving the goal. Requires search and planning.
Example: Navigation system finding route to a destination

Utility-Based Agent

Assigns a utility score to states. Chooses actions that maximise expected utility. Handles conflicting goals and uncertainty.

Example: Self-driving car balancing speed, safety, and comfort

Learning Agent

Improves its own performance over time. Has four components: Performance Element, Learning Element, Critic, Problem Generator.

Example: AlphaGo learning Go by playing millions of games

9. Learning Agent — Component Diagram

Learning Element	Performance Element	Critic	Problem Generator
Improves the performance element	Selects actions to take	Evaluates performance against standard	Suggests exploratory new actions

APPLICATIONS & FUTURE OF AI

10. Current Applications of AI

Domain	Example Application	AI Technique Used
Autonomous Vehicles	Waymo self-driving car, drone delivery	Computer vision, planning, deep RL
Game Playing	AlphaGo, AlphaZero, Dota 2, StarCraft II	Deep reinforcement learning, MCTS
Language & Speech	Siri, Alexa, Google Translate (100+ languages)	NLP, transformers, LSTMs
Medicine	Cancer detection, diabetic retinopathy screening	CNNs, deep learning on medical images
Recommendations	Netflix, Amazon, Spotify, spam filters	Collaborative filtering, ML
Image Understanding	Face unlock, Google Lens, image captioning	Convolutional neural networks
Planning & Scheduling	NASA Remote Agent, EUROPA, DART military logistics	Search, constraint satisfaction

Domain	Example Application	AI Technique Used
Climate Science	Predicting extreme weather events	Deep learning on satellite data

11. Future of AI — Benefits & Risks

✓Benefits	⍰ Risks
• Frees humans from menial/repetitive work • Accelerates scientific research • Improves healthcare & diagnostics • Better resource management • Climate change solutions • Increased productivity	• Lethal autonomous weapons • Mass surveillance & persuasion • Biased decision-making (race, gender) • Job displacement • Safety-critical system failures • Cybersecurity threats

11.1 Long-Term Concerns

Concern	What it Means
AGI (Human-Level AI)	A machine that can do anything a human can
ASI (Superintelligence)	Intelligence far surpassing the best human minds
Gorilla Problem	Just as gorillas depend on humans, humans may depend on/be controlled by AI
King Midas Problem	AI achieves exactly what we asked but not what we wanted
Value Alignment Problem	Ensuring machine objectives truly match human values
Control Problem	How do we control machines more intelligent than us?

QUICK REVISION — CHAPTER 1 & 2 SUMMARY

Chapter 1 vs. Chapter 2 — At a Glance

Aspect	Chapter 1	Chapter 2
Focus	What AI is, history, foundations, applications, risks	Agent design, environments, rationality, structures
Key Concepts	Four approaches, Turing Test, Standard Model, Beneficial AI	Agent, Percept, Rational Agent, PEAS, Env. properties, Agent types

Aspect	Chapter 1	Chapter 2
Main Contribution	Broad overview and motivation for studying AI	Technical framework for building rational agents

Key Points to Remember

- AI has 4 approaches: Acting Humanly, Thinking Humanly, Thinking Rationally, Acting Rationally
- The Rational Agent approach is the standard modern model of AI
- PEAS = Performance, Environment, Actuators, Sensors
- Agent = Architecture + Program
- 5 Agent Types: Simple Reflex → Model-Based → Goal-Based → Utility-Based → Learning
- Learning Agent has 4 parts: Learning Element, Performance Element, Critic, Problem Generator
- Hardest environment: partially observable + multiagent + stochastic + sequential + dynamic + continuous + unknown
- Value Alignment = making sure machine goals match human values
- Future risks include weapons, surveillance, bias, job loss, safety, and cybersecurity
- Long-term: AGI, ASI, Gorilla problem, King Midas problem, Control problem

Exam Tip: Always relate agent types to their environments. A simple reflex agent fails in partial observability — a model-based agent handles it!